

MYSTERIOUS PATTERNS REVEALED BY "SYSTEMS"

This column introduces a variety of physical phenomena and fascinating scientific concepts.

In this memorable first column, we will explain the concept of attractors—specifically, “strange attractors”, which are characterized by their intriguing shapes.

ATTRACTORS

This refers to the stable state attained by a dissipative system—a system involving the exchange of energy (i.e., a non-conservative system)—after a sufficient amount of time has elapsed, wherein the system settles into a specific trajectory or point. Fundamentally, attractors are classified into four types: equilibrium point attractors, periodic attractors, quasi-periodic attractors, and strange attractors.

For example, a system in which a pendulum eventually comes to rest at a single point is classified as a fixed-point attractor.

STRANGE ATTRACTOR

A famous example is the Lorenz attractor. It is well known for its resemblance to a butterfly's wings; while its trajectory oscillates back and forth between the left and right sides, it never traces out the exact same path!

A strange attractor is a type of attractor—a collection of complex trajectories that, while never repeating themselves exactly, do not simply scatter indefinitely.

The figure on the right depicts "THOMAS' CYCLICALLY SYMMETRIC ATTRACTORS," created using TouchDesigner, a node-based programming environment. By varying the value of $*b*$ (a constant), you can observe the attractor taking on a variety of distinct forms. The equations used to generate this figure are as follows:

$$\begin{aligned}dx/dt &= \sin(y) - bx \\ dy/dt &= \sin(z) - by \\ dz/dt &= \sin(x) - bz\end{aligned}$$



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